

Sessional Exam 1
Introduction to Data Science and Design Thinking (BCE-O633)

Note: Section A contains 4 questions attempt any 2 (5 marks Each), Section B contains 2 questions, attempt any 1 (10 marks each)

Section A

- Q1. Define the term Data Science, and explain steps to learn data science.
- Q2. Discuss the 5 v's in terms of data.
- Q3. What are the stages in a data science project? explain each phase in brief.
- Q4. What are data security issues in a data science project?

Section B

- Q1. You own a bike share company and you want to increase your revenue in next semester. What will be your policy to achieve the goal? Discuss your approach with respect to different Data Science Roles and explain the task of each persona.
- Q2. What are the applications of Data Science in various fields? Mention any five applications with detail.

Sessional Exam 2
Introduction to Data Science and Design Thinking (BCE-O633)

Note: Section A contains 4 questions attempt any 2 (5 marks Each), Section B contains 2 questions, attempt any 1 (10 marks each)

Section A

Q1. Define Data Integration and its challenges.

Q2. Discuss data collection strategies. Mention any five data collection methods.

Q3. Find mean, median and mode for the following data

34, 45, 23, 87, 96, 34, 56, 78, 23, 11, 22, 23, 101, 124

Q4. Create a box plot for the above data. Find all five elements of box plot.

Section B

Q1. Define data pre-processing and its major task and also discuss why it is required.

Q2. Discuss data transformation strategies, define any five strategies in detail.